Gibbs Sampling

A Mathematical Derivation for Graduate Students in the Physical Sciences and Mathematics

Morten Hjorth-Jensen

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Goals of These Notes

These notes develop Gibbs sampling from a mathematical viewpoint, with emphasis on:

- conditional distributions,
- Markov chains and invariant measures,
- detailed balance,
- convergence intuition,
- and connections to statistical physics and Bayesian inference.

The aim is to explain not only *how* Gibbs sampling works, but *why* it correctly samples from a target distribution.

The Basic Sampling Problem

Suppose we want to sample from a probability distribution

$$\pi(x_1, \ldots, x_d)$$

on a high-dimensional space.

Typical situations:

- a Boltzmann distribution in statistical mechanics,
- a posterior distribution in Bayesian inference,
- a latent-variable model in machine learning.

In many important cases:

- π is known only up to normalization,
- direct sampling is difficult,
- but conditional distributions are simple.

This is exactly the setting where Gibbs sampling becomes useful.

Target Distribution and Normalization

Often the target distribution has the form

$$\pi(x) = \frac{1}{Z} f(x),$$

where

$$Z = \int f(x)\, dx$$

or, in the discrete case,

$$Z = \sum_x f(x).$$

In statistical physics one often writes

$$\pi(x) = \frac{1}{Z} e^{-\beta E(x)},$$

where:

- $E(x)$ is an energy,
- $\beta = 1/(k_B T)$,
- Z is the partition function.

Why Conditional Distributions Help

Even when $\pi(x_1, \ldots, x_d)$ is hard to sample from directly, the conditional distributions

$$\pi(x_i \mid x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_d)$$

may be easy to sample.

The core idea of Gibbs sampling is:

> *sample one coordinate at a time from its exact conditional distribution, while keeping the others fixed.*

This creates a Markov chain whose stationary distribution is the desired target π.

Conditional Probability

For two variables X, Y, the conditional distribution is

$$\pi(x \mid y) = \frac{\pi(x, y)}{\pi_Y(y)}, \qquad \pi_Y(y) = \int \pi(x, y)\, dx,$$

provided $\pi_Y(y) > 0$.
In d dimensions, the conditional distribution for coordinate i is

$$\pi(x_i \mid x_{-i}) = \frac{\pi(x_i, x_{-i})}{\int \pi(x_i', x_{-i})\, dx_i'}.$$

Here x_{-i} means all coordinates except x_i.

The Gibbs Sampling Update Rule

Let the current state be

$$x^{(t)} = (x_1^{(t)}, \ldots, x_d^{(t)}).$$

One full Gibbs sweep updates coordinates sequentially:

$$x_1^{(t+1)} \sim \pi(x_1 \mid x_2^{(t)}, \ldots, x_d^{(t)}),$$

$$x_2^{(t+1)} \sim \pi(x_2 \mid x_1^{(t+1)}, x_3^{(t)}, \ldots, x_d^{(t)}),$$

and so on until

$$x_d^{(t+1)} \sim \pi(x_d \mid x_1^{(t+1)}, \ldots, x_{d-1}^{(t+1)}).$$

This defines a Markov chain on the state space.

Two-Variable Example

Suppose the target is $\pi(x, y)$.
A Gibbs iteration is:

$$x^{(t+1)} \sim \pi(x \mid y^{(t)}),$$
$$y^{(t+1)} \sim \pi(y \mid x^{(t+1)}).$$

This is already enough to understand the general structure:

- each update is exact conditional sampling,
- the resulting process is Markovian,
- the full joint distribution is preserved.

Transition Kernel for One Coordinate Update

Fix a coordinate i. A single-coordinate Gibbs update defines a transition kernel

$$K_i(x, dx') = \delta_{x_{-i}}(dx'_{-i})\, \pi(dx'_i \mid x_{-i}).$$

Interpretation:

- all coordinates except i remain unchanged,
- coordinate i is redrawn from its conditional law.

A full Gibbs sweep is then the composition

$$K = K_d \cdots K_2 K_1.$$

Invariant Distribution: Strategy

We want to show that π is invariant under Gibbs updates.

That means:

$$\int \pi(dx)\, K(x, A) = \pi(A)$$

for every measurable set A.
It is enough to prove this first for a single-coordinate update K_i, then invariance for the full sweep follows by composition.

Invariance of a Single-Coordinate Update

Consider one coordinate i. Then

$$\int \pi(dx)\, K_i(x, dx') = \int \pi(dx_i, dx_{-i})\, \delta_{x_{-i}}(dx'_{-i})\, \pi(dx'_i \mid x_{-i}).$$

Using the factorization

$$\pi(dx_i, dx_{-i}) = \pi(dx_i \mid x_{-i})\, \pi_{-i}(dx_{-i}),$$

we integrate over x_i:

$$\int \pi(dx_i \mid x_{-i}) = 1.$$

Hence

$$\int \pi(dx)\, K_i(x, dx') = \pi(dx'_i \mid x'_{-i})\, \pi_{-i}(dx'_{-i}) = \pi(dx').$$

So π is invariant for each coordinate update.

Detailed Balance for a Single Update

In fact, each single-coordinate Gibbs update satisfies detailed balance with respect to π:

$$\pi(dx)\, K_i(x, dx') = \pi(dx')\, K_i(x', dx).$$

Why?

Since only coordinate i changes, we must have $x_{-i} = x'_{-i}$. Then

$$\pi(x)\, K_i(x, x') = \pi(x_i, x_{-i})\, \pi(x'_i \mid x_{-i}),$$

and similarly

$$\pi(x')\, K_i(x', x) = \pi(x'_i, x_{-i})\, \pi(x_i \mid x_{-i}).$$

But using

$$\pi(x_i, x_{-i}) = \pi(x_i \mid x_{-i})\pi(x_{-i}),$$

both sides become

$$\pi(x_{-i})\, \pi(x_i \mid x_{-i})\, \pi(x'_i \mid x_{-i}).$$

Why Detailed Balance Matters

Detailed balance implies invariance:

$$\pi(dx)\,K(x, dx') = \pi(dx')\,K(x', dx) \quad \Longrightarrow \quad \pi K = \pi.$$

This gives a simple physical picture:

in equilibrium, probability flux from x to x' is exactly balanced by the reverse flux.

For Gibbs sampling:

- each coordinate update is reversible,
- the full deterministic sweep is invariant,
- random-scan Gibbs is also reversible under suitable conditions.

Random-Scan Gibbs Sampling

Instead of updating coordinates in fixed order, one may choose a random coordinate $I_t \in \{1, \ldots, d\}$ at each step, for example uniformly:

$$\mathbb{P}(I_t = i) = \frac{1}{d}.$$

Then update

$$x_i^{(t+1)} \sim \pi(x_i \mid x_{-i}^{(t)}),$$

while leaving the remaining coordinates unchanged.
The transition kernel becomes

$$K = \frac{1}{d} \sum_{i=1}^{d} K_i.$$

Since each K_i leaves π invariant, so does K.

Gibbs Sampling as a Special Case of Metropolis–Hastings

Metropolis–Hastings proposes a move $x \to x'$ and accepts it with probability

$$\alpha(x, x') = \min\left(1, \frac{\pi(x')q(x \mid x')}{\pi(x)q(x' \mid x)}\right).$$

In Gibbs sampling, the proposal is exactly the conditional distribution:

$$q(x_i' \mid x) = \pi(x_i' \mid x_{-i}).$$

Then the acceptance ratio becomes

$$\frac{\pi(x')q(x \mid x')}{\pi(x)q(x' \mid x)} = 1,$$

so every move is accepted.
Thus Gibbs sampling is a rejection-free special case of Metropolis–Hastings.

Connection to the Boltzmann Distribution

Suppose

$$\pi(x) = \frac{1}{Z} e^{-\beta E(x)}.$$

Then the conditional distribution for coordinate x_i is

$$\pi(x_i \mid x_{-i}) = \frac{e^{-\beta E(x_i, x_{-i})}}{\sum_{x_i'} e^{-\beta E(x_i', x_{-i})}}$$

in the discrete case.
This is the local Boltzmann law: update one degree of freedom while the environment x_{-i} is frozen.

Example: Ising Model

For Ising spins $s_i \in \{-1, +1\}$, the energy is

$$E(s) = -\sum_{\langle i,j \rangle} J_{ij} s_i s_j - \sum_i h_i s_i.$$

The Boltzmann distribution is

$$\pi(s) = \frac{1}{Z} e^{-\beta E(s)}.$$

To update spin s_i, only terms involving s_i matter:

$$E(s) = -s_i \left(\sum_{j \neq i} J_{ij} s_j + h_i \right) + \text{const.}$$

Hence

$$\pi(s_i \mid s_{-i}) \propto \exp \left[\beta s_i \left(\sum_{j \neq i} J_{ij} s_j + h_i \right) \right].$$

Explicit Conditional for the Ising Spin

Define the local field

$$H_i^{\mathrm{loc}} = \sum_{j \neq i} J_{ij} s_j + h_i.$$

Then

$$\mathbb{P}(s_i = +1 \mid s_{-i}) = \frac{e^{\beta H_i^{\mathrm{loc}}}}{e^{\beta H_i^{\mathrm{loc}}} + e^{-\beta H_i^{\mathrm{loc}}}} = \frac{1}{1 + e^{-2\beta H_i^{\mathrm{loc}}}}.$$

Similarly,

$$\mathbb{P}(s_i = -1 \mid s_{-i}) = \frac{1}{1 + e^{2\beta H_i^{\mathrm{loc}}}}.$$

This logistic form is one of the most important explicit examples of Gibbs sampling in physics.

Continuous Example: Bivariate Gaussian

Let

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix} \right).$$

Then the conditional distributions are Gaussian:

$$X \mid Y = y \sim \mathcal{N}\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y),\ \sigma_X^2(1-\rho^2) \right),$$

$$Y \mid X = x \sim \mathcal{N}\left(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X),\ \sigma_Y^2(1-\rho^2) \right).$$

So Gibbs sampling alternates exact Gaussian draws.

Convergence: What Must Be True?

Invariance alone is not enough. We also need the Markov chain to converge to π.
Typical sufficient conditions are:

- irreducibility,
- aperiodicity,
- positive recurrence.

Under such conditions, Gibbs sampling satisfies

$$\lim_{t \to \infty} \mathbb{P}(X^{(t)} \in A) = \pi(A)$$

for suitable initial conditions.
So the chain forgets its initial state and approaches equilibrium.

Ergodic Theorem

If the Gibbs chain is ergodic and f is integrable, then

$$\frac{1}{N}\sum_{t=1}^{N} f(X^{(t)}) \longrightarrow \mathbb{E}_{\pi}[f]$$

almost surely as $N \to \infty$.
This is the practical reason Gibbs sampling is useful: we can estimate expectations under π by time averages along the chain.

Examples:

- magnetization in the Ising model,
- posterior means in Bayesian inference,
- correlation functions in lattice models.

Why Gibbs Sampling Can Be Slow

Even though the algorithm is simple, convergence may be slow if:

- variables are strongly correlated,
- the target distribution is multimodal,
- local updates move only small distances,
- there is critical slowing down near phase transitions.

This is especially important in statistical physics, where local Gibbs updates may decorrelate very slowly near criticality.

Blocked Gibbs Sampling

A standard improvement is to update blocks of variables together.
Instead of sampling one coordinate x_i, sample a block x_B:

$$x_B^{(t+1)} \sim \pi(x_B \mid x_{-B}^{(t)}).$$

This can reduce autocorrelation when variables inside the block are strongly coupled.

So the Gibbs philosophy remains the same:

sample exactly from conditional distributions

but the conditionals are taken for larger subsets.

Mathematical Summary of the Derivation

The logic of Gibbs sampling is:

1. start from a target distribution $\pi(x_1, \ldots, x_d)$,
2. identify tractable conditional laws $\pi(x_i \mid x_{-i})$,
3. define a Markov chain by sequentially sampling from these conditionals,
4. prove that each coordinate update preserves π,
5. conclude that π is invariant for the full chain,
6. under ergodicity conditions, conclude convergence to π.

Physical Interpretation

In statistical physics, Gibbs sampling can be interpreted as:

- local equilibration of one degree of freedom,
- while the surrounding environment is held fixed,
- repeated many times until global equilibrium is reached.

This makes it a natural algorithm for:

- spin systems,
- lattice field models,
- Boltzmann machines,
- Bayesian graphical models.

Final Take-Home Message

Gibbs sampling is mathematically simple but conceptually deep.

It works because:

- conditional distributions are exact,
- the induced Markov chain preserves the full target distribution,
- detailed balance and invariance guarantee correctness,
- ergodicity guarantees convergence.

So Gibbs sampling is one of the clearest examples of how local probabilistic updates can reconstruct a global equilibrium distribution.